Variational Quantum Circuits for Pesticide Environmental Fate Prediction: A Comparative Study with Classical Machine Learning on the EU SPIN Database

Celal Arda

Independent Researcher

celal.arda@outlook.de

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Abstract

Predicting pesticide environmental fate properties—specifically soil degradation half-life (DegT50) and organic carbon sorption coefficient (K_{oc})—is essential for regulatory risk assessment under EC 1107/2009. Classical QSAR models achieve moderate predictive power for K_{oc} but struggle with DegT50 due to its dependence on complex soil microbial processes. This work investigates whether variational quantum circuits (VQCs) can capture non-linear molecular descriptor relationships that elude classical regressors. We encode 21 physicochemical descriptors for 110 EU-registered pesticides into per-target quantum circuits: an 8-qubit circuit for DegT50 (17 core features) and a 12-qubit circuit for K_{oc} (all 21 features), each with IQP-style encoding and data re-uploading. Leave-one-out cross-validation benchmarks reveal that Random Forest (RF) achieves R^2 = 0.287 for DegT50 and 0.759 for K_{oc}, while Gradient Boosting (GB) reaches 0.283 and 0.752, respectively. An MLP neural network achieves only 0.075 and 0.497. The VQC alone underperforms ($R^2 = -0.028$ for DegT50, 5-fold CV), but a hybrid stacking framework blending QML and RF predictions yields $\alpha = 0.30$, achieving $R^2 = 0.231$ for DegT50 (+19% over standalone RF). This demonstrates that VQCs capture complementary molecular patterns not accessible to tree-based models. These results provide a rigorous benchmark for near-term quantum machine learning applied to regulatory chemistry and highlight the critical role of per-target circuit design and hybrid stacking in small-dataset QML.

Keywords: quantum machine learning, variational quantum circuits, pesticide fate, DegT50, Koc, QSAR, hybrid stacking, regulatory chemistry, EU SPIN

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1 Introduction

Pesticide environmental fate assessment is a cornerstone of EU regulatory approval under Regulation (EC) No 1107/2009 (ec1, 2009). Two properties dominate the risk evaluation: the soil degradation half-life (DegT50), governing persistence, and the organic carbon sorption coefficient (K_{oc}), governing mobility and leaching potential. Together, these parameters feed into the FOCUS groundwater models (foc, 2006) that determine whether a substance can be approved for agricultural use.

Experimental determination of DegT50 follows OECD Guideline 307 (oec, 2002), requiring months-long incubation studies under controlled temperature and moisture. K_{oc} is measured via batch equilibrium adsorption (OECD 106) (oec, 2000). Both are expensive, time-consuming, and subject to substantial inter-laboratory variability. In silico prediction of these properties from molecular structure alone would accelerate the regulatory pipeline and reduce animal testing requirements.

Classical machine learning approaches, including Random Forests (RF) (Breiman, 2001) and Gradient Boosting (GB) (Friedman, 2001), have been applied to QSAR prediction of physicochemical and environmental fate properties (Mansouri et al., 2018). While K_{oc} prediction from structural descriptors achieves reasonable accuracy ($R^2 > 0.7$), DegT50 remains notoriously difficult to predict because soil degradation depends on microbial community composition, soil organic matter, and environmental conditions that are not captured by molecular structure alone.

Quantum machine learning (QML) offers a fundamentally different computational paradigm. Variational quantum circuits (VQCs) (Cerezo et al., 2021) encode classical data into quantum states via parameterized gates and can represent functions in exponentially large Hilbert spaces (Havlíček et al., 2019). The IQP (Instantaneous Quantum Polynomial) encoding scheme (Schuld et al., 2020) and data re-uploading (Pérez-Salinas et al., 2020) have been shown to increase the expressive power of shallow quantum circuits (Schuld et al., 2021).

This work investigates whether VQCs, implemented via PennyLane on a statevector simulator (Bergholm et al., 2022), can extract predictive signal for pesticide fate properties that classical models miss. Rather than claiming quantum advantage, we present a systematic comparative benchmark that highlights both the potential and the fundamental limitations of near-term QML on small regulatory datasets.

2 Materials and Methods

2.1 Dataset

The dataset comprises 110 pesticide active substances registered in the EU Substances, Preparations and Indicators Navigator (EU SPIN) (eus, 2024). These span 50 chemical classes in-

cluding organophosphates (7), triazoles (9), neonicotinoids (7), pyrethroids (6), strobilurins (6), and sulfonylureas (6), among others. Target properties include:

- DegT50 (soil): laboratory degradation half-life in days, range 0.5 d to 1389 d (median 25 d, mean 75.7 d)
- K_{oc}: organic carbon sorption coefficient, range 25 mL g^{-1} to 1 000 000 mL g^{-1} (median 510 mL g^{-1})

Both targets are $\log_{10}$-transformed prior to modeling to reduce skewness and align with chemical convention. The target distributions are shown in fig. 1.

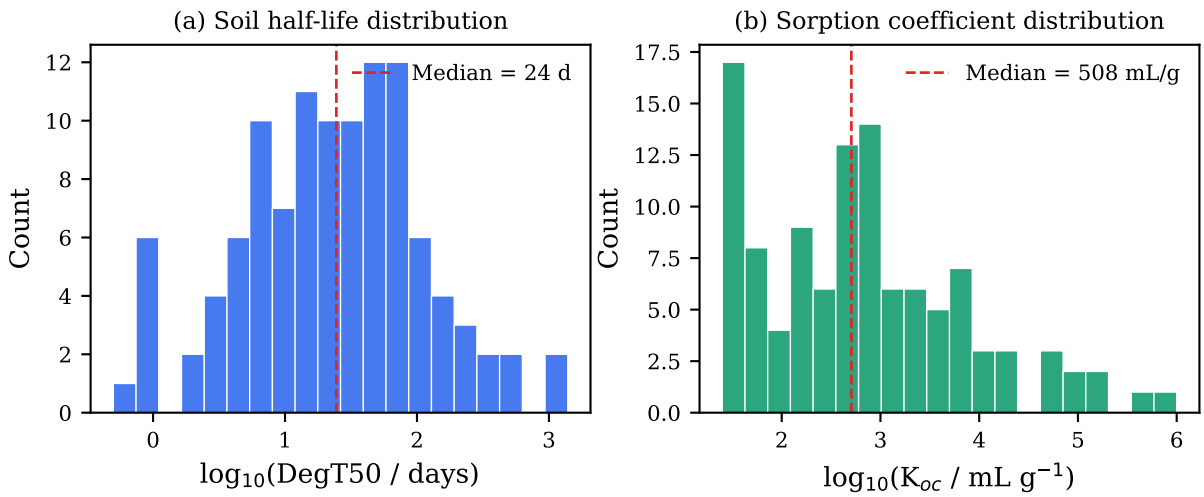

Figure 1: Distribution of (a) soil degradation half-life (DegT50) and (b) organic carbon sorption coefficient (K_{oc}) for the 110 EU SPIN pesticides, shown on $\log_{10}$ scale. Red dashed lines indicate medians.

2.2 Molecular Descriptor Features

We compute 21 molecular descriptor features for each substance, organized in four phases of development:

1. **Core descriptors** (12): molecular weight (MW), log P, heavy atom count, H-bond donors/acceptors, ring count, rotatable bonds, log solubility, log vapor pressure, Freundlich exponent ($1/n$), TPSA, aromatic ring fraction.
2. **Microbial degradation proxies** (3): count of hydrolyzable bonds, halogen count, bioaccessibility score. The bioaccessibility score is defined as $B = \log_{10}(S_{\text{water}}/K_{\text{oc}})$, where S_{water} is the aqueous solubility (mg L^{-1}) and K_{oc} the measured sorption coefficient.
   Circularity note: Because B includes the measured K_{oc}, using it as a feature when *predicting* K_{oc} introduces partial data leakage. For DegT50 prediction this is unproblematic (the target is unrelated to K_{oc}), but for K_{oc} prediction we report all results both with the full feature set and with reduced sets that exclude B (table 1).
3. **Blind-spot corrections** (2): charge state at soil pH 6.5, conjugated π-system size.
4. **DegT50-targeted descriptors** (4): heteroatom ratio (N/O/S/P density as a proxy for microbial attackability), sp^3 fraction (molecular shape and flexibility), Henry-to-solubility

ratio (volatilization pathway proxy), and oxidation susceptibility (count of cytochrome P450 metabolic lability sites: tertiary C–H, benzylic, allylic, thioether, and secondary amine substructures). These four features were added in Phase 5c based on residual error analysis of the worst-predicted substances.

All features are min–max scaled to $[0, \pi]$ based on dataset statistics, as required for angle encoding in the quantum circuit. Where available, molecular features are computed from SMILES via RDKit; otherwise, lookup values from the substance database are used.

The Random Forest feature importances (fig. 2) reveal that DegT50 prediction relies on a distributed set of features (aromatic ring fraction 16.3%, Freundlich $1/n$ 13.2%, $\log P$ 8.7%), while K_{oc} is dominated by bioaccessibility (52.3%) and $\log P$ (19.7%).

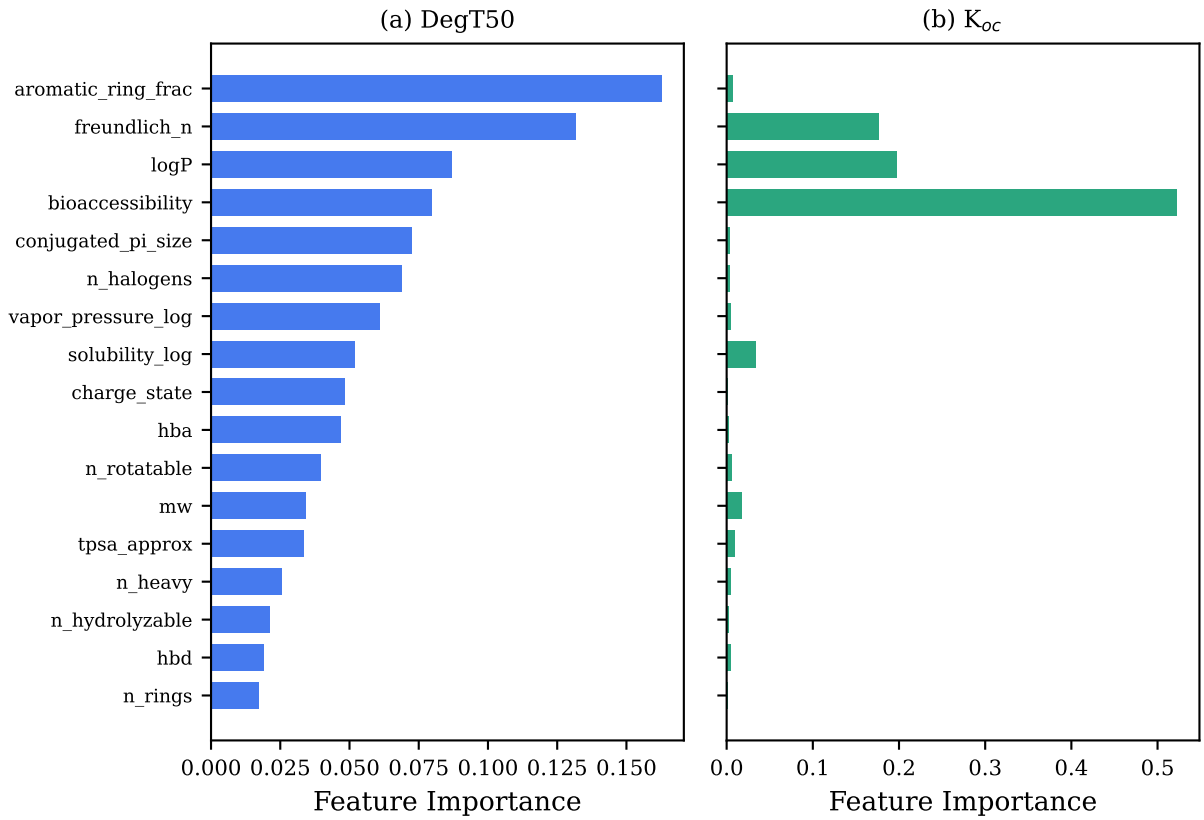

Figure 2: Random Forest feature importances for (a) DegT50 and (b) K_{oc} prediction, trained on the full 110-substance dataset. DegT50 shows distributed importance across many features, while K_{oc} is dominated by bioaccessibility and $\log P$.

2.3 Variational Quantum Circuit Architecture

We employ per-target circuits optimized for each prediction task: an 8-qubit, 6-layer circuit for DegT50 (using the 17-feature core set) and a 12-qubit, 8-layer circuit for K_{oc} (using all 21 features). Both circuits share the same architecture (fig. 3):

1. **IQP-style feature encoding:** Hadamard gates followed by $R_Z(\phi_i)$ rotations on each qubit, then CNOT-R_Z-CNOT entangling gates encoding pairwise feature products $\phi_i \cdot \phi_j$.

2. **Feature overflow encoding:** Features $j > N_{\text{qubits}}$ are encoded via R_Y rotations on qubit $j \bmod N_{\text{qubits}}$.
3. **Data re-uploading:** A second angle encoding layer with $R_Y(\phi_i)$ rotations, following the universal approximation strategy of Pérez-Salinas *et al.* (Pérez-Salinas et al., 2020).
4. **Variational layers:** L layers of single-qubit $\text{Rot}(\theta, \phi, \omega)$ rotations and alternating nearest-neighbor CNOT entangling patterns.
5. **Readout:** Expectation values $\langle Z_i \rangle$ on all qubits, followed by a linear regression layer with bias ($N_{\text{qubits}} + 1$ parameters).

For the DegT50 circuit (8q/6L): $6 \times 8 \times 3 = 144$ variational plus 9 readout = 153 parameters ($N_{\text{params}}/N_{\text{data}} = 1.39$). For the K_{oc} circuit (12q/8L): $8 \times 12 \times 3 = 288$ variational plus 13 readout = 301 parameters ($N_{\text{params}}/N_{\text{data}} = 2.74$). The per-target design balances expressivity against overfitting risk:

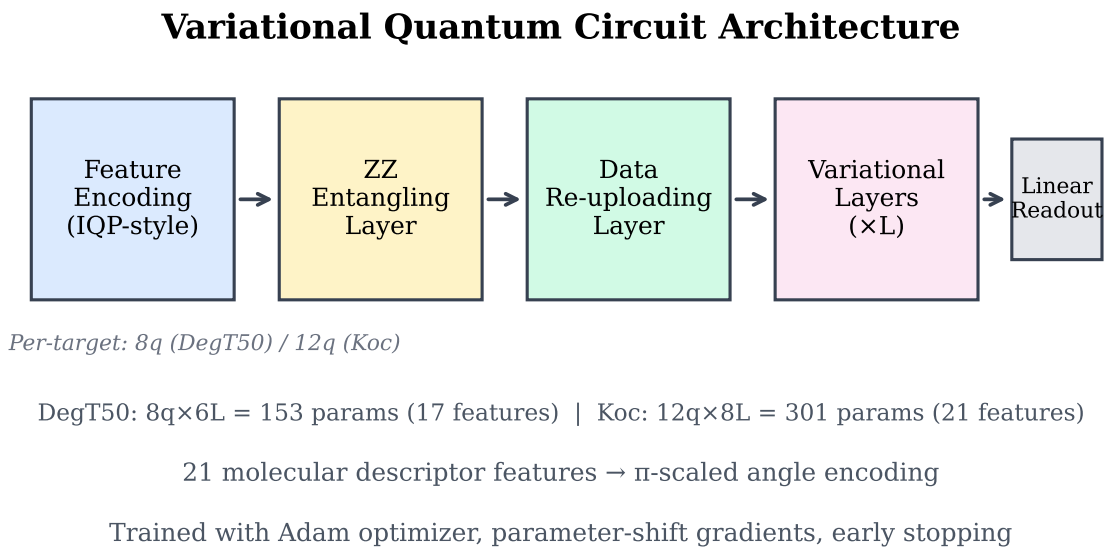

Figure 3: Schematic of the variational quantum circuit architecture. Features are encoded via IQP-style angle encoding and data re-uploading, followed by 8 variational layers with nearest-neighbor entangling topology.

2.4 Training Procedure

Separate circuits are trained for DegT50 and K_{oc} prediction. Variational parameters are initialized uniformly in $[-0.5, 0.5]$ with a fixed random seed (42) for reproducibility. Optimization uses the Adam optimizer (Kingma and Ba, 2015) with learning rate 0.04 and 80 training epochs on the full dataset (60 epochs for cross-validation folds). Early stopping with patience 10 is applied: training halts if the MSE does not improve for 10 consecutive epochs, mitigating overfitting on the smaller DegT50 circuit. Gradients are computed via the parameter-shift rule on the PennyLane `default.qubit` statevector simulator.

Critically, the DegT50 QML circuit receives only the 17 core features (indices 0–16), while the K_{oc} circuit receives all 21 features. This per-model feature selection was motivated by empirical observation that the four Phase 5c features improve RF performance but degrade QML performance due to the increased information compression into limited qubit capacity.

2.5 Classical Baselines

Four classical regressors are trained on the identical feature sets using scikit-learn (Pedregosa et al., 2011):

- **Ridge Regression:** L_2-penalized linear model ($\alpha = 1.0$) (Hoerl and Kennard, 1970).
- **LASSO:** L_1-penalized linear model ($\alpha = 0.1$) (Tibshirani, 1996).
- **Random Forest (RF):** 200 trees, max depth 10.
- **Gradient Boosting (GB):** 200 estimators, max depth 4, learning rate 0.1.
- **MLP Neural Network:** Two hidden layers (64, 32 neurons), ReLU activation, Adam optimizer, early stopping with 15% validation split, L_2 regularization ($\alpha = 0.01$). Features are standardized per fold.

All models use the same random seed (42) for reproducibility.

2.6 Cross-Validation Strategy

Leave-one-out cross-validation (LOO-CV) is the primary evaluation metric, as it maximizes training data usage for the small 110-substance dataset. Each fold trains the model on $N-1$ substances and predicts the held-out substance. 5-fold CV with stratified random splits is used as a faster supplementary benchmark.

For the VQC, LOO-CV requires 110×2 (two properties) full training runs, each with 60 epochs of gradient optimization over 153–301 parameters. Fold-level checkpointing saves results after each fold to a JSON file, enabling crash-resilient resumption.

2.7 Hybrid QML+RF Stacking

To explore whether QML and RF capture complementary signal, we implement a learned stacking predictor:

$$\hat{y}_{\text{hybrid}} = \alpha \cdot \hat{y}_{\text{QML}} + (1-\alpha) \cdot \hat{y}_{\text{RF}} \tag{1}$$

where $\alpha \in [0,1]$ is optimized via grid search (step 0.05) to minimize the LOO-CV mean squared error. If $\alpha = 0$, the QML component adds no value; if $\alpha > 0$, the VQC captures complementary information.

2.8 Adverse Outcome Pathway Features

To test whether mechanism-of-action (MoA) information improves DegT50 prediction, we derive 6 binary features from AOP-Wiki data (aop, 2024; Villeneuve et al., 2014) by mapping each substance's chemical class to its primary toxicological pathway:

1. AChE inhibition (organophosphates, carbamates; 15 substances)
2. GABA/ion channel disruption (pyrethroids; 11 substances)
3. Mitochondrial respiration (SDHI, strobilurins; 12 substances)
4. Nicotinic AChR agonism (neonicotinoids; 10 substances)
5. Endocrine disruption (triazines; 13 substances)
6. Cell division inhibition (benzimidazoles; 6 substances)

Coverage: 67/110 substances (60.4%) are assigned to at least one pathway category (fig. 7).

3 Results

3.1 Classical Baseline Performance

Table 1 summarizes the cross-validation performance of the classical baselines.

Key observations:

- **Bioaccessibility inflates linear K_{oc} models:** Ridge R^2 drops from 0.842 to 0.546 ($\Delta = -0.30$) when the circular bioaccessibility feature is removed, because B is a near-linear transform of the target. Ensemble models are far more robust: GB loses only 0.015 in LOO-CV R^2 ($0.779 \rightarrow 0.764$), because tree-based methods can derive the same information from $\log P$ and solubility independently.
- **GB leads fair K_{oc} prediction:** On the circularity-free 16-feature set, Gradient Boosting achieves the highest LOO-CV R^2 (0.764), followed by RF (0.726), Ridge (0.546), and LASSO (0.535). The nonlinear advantage of tree-based models is now evident: K_{oc} prediction benefits from interaction effects that linear models cannot capture without the B shortcut.
- DegT50 prediction is poor across all models ($R^2 < 0.25$), with linear and nonlinear methods performing comparably. The ensemble models' modest advantage (GB 0.223 vs. Ridge 0.154) suggests limited nonlinear signal for soil degradation from structural descriptors alone.
- **DegT50 instability:** The fold-level standard deviations for DegT50 5-fold CV ($\pm$0.10–0.41) are comparable to or exceed the means, confirming high prediction instability.

Wilcoxon signed-rank tests on per-substance LOO squared residuals (17-feature set) confirm that none of the pairwise model differences are statistically significant at $\alpha = 0.05$: RF vs. GB ($p = 0.19$ for DegT50, $p = 0.21$ for K_{oc}), Ridge vs. RF ($p = 0.16$, $p = 0.07$), and Ridge vs. GB ($p = 0.71$, $p = 0.18$).

Table 1: Baseline performance (leave-one-out and 5-fold CV). All metrics computed on $\log_{10}$-scaled targets; 5-fold values report per-fold mean R^2. K_{oc} results are shown both with the full 17-feature set (including bioaccessibility B) and the fair 16-feature set (excluding B; see circularity note in section 2.2).

Model	**Property**	**CV**	$\mathbf{R^2}$	**MAE**	**RMSE**
Ridge	DegT50	LOO	0.154	0.488	0.626
	DegT50	5-fold	−0.109	0.504	0.644
	K_{oc} (17f)	LOO	$0.842^{\dagger}$	0.252	0.413
	K_{oc} (16f)	LOO	0.546	0.439	0.699
	K_{oc} (17f)	5-fold	$0.848^{\dagger}$	0.248	0.405
	K_{oc} (16f)	5-fold	0.528	0.441	0.698
LASSO	DegT50	LOO	0.101	0.518	0.645
	DegT50	5-fold	−0.050	0.525	0.658
	K_{oc} (17f)	LOO	$0.747^{\dagger}$	0.336	0.522
	K_{oc} (16f)	LOO	0.535	0.476	0.708
	K_{oc} (17f)	5-fold	$0.740^{\dagger}$	0.341	0.529
	K_{oc} (16f)	5-fold	0.528	0.471	0.706
Random Forest	DegT50	LOO	0.194	0.474	0.611
	DegT50	5-fold	0.089	0.464	0.596
	K_{oc} (17f)	LOO	0.766	0.304	0.502
	K_{oc} (16f)	LOO	0.726	0.341	0.543
	K_{oc} (17f)	5-fold	0.766	0.299	0.502
	K_{oc} (16f)	5-fold	0.726	0.346	0.552
Gradient Boosting	DegT50	LOO	0.223	0.487	0.600
	DegT50	5-fold	0.034	0.470	0.606
	K_{oc} (17f)	LOO	0.779	0.288	0.488
	K_{oc} (16f)	LOO	0.764	0.323	0.504
	K_{oc} (17f)	5-fold	0.780	0.292	0.487
	K_{oc} (16f)	5-fold	0.740	0.341	0.542

†Inflated by bioaccessibility circularity: $B = \log_{10}(S/K_{oc})$ uses measured K_{oc} as input. The 16-feature (16f) rows exclude B and represent the fair comparison. 17f = all 17 features including B.

The predicted-vs-experimental plots for RF LOO-CV (fig. 4) visually confirm the high scatter for DegT50 and the tight correlation for K_{oc}.

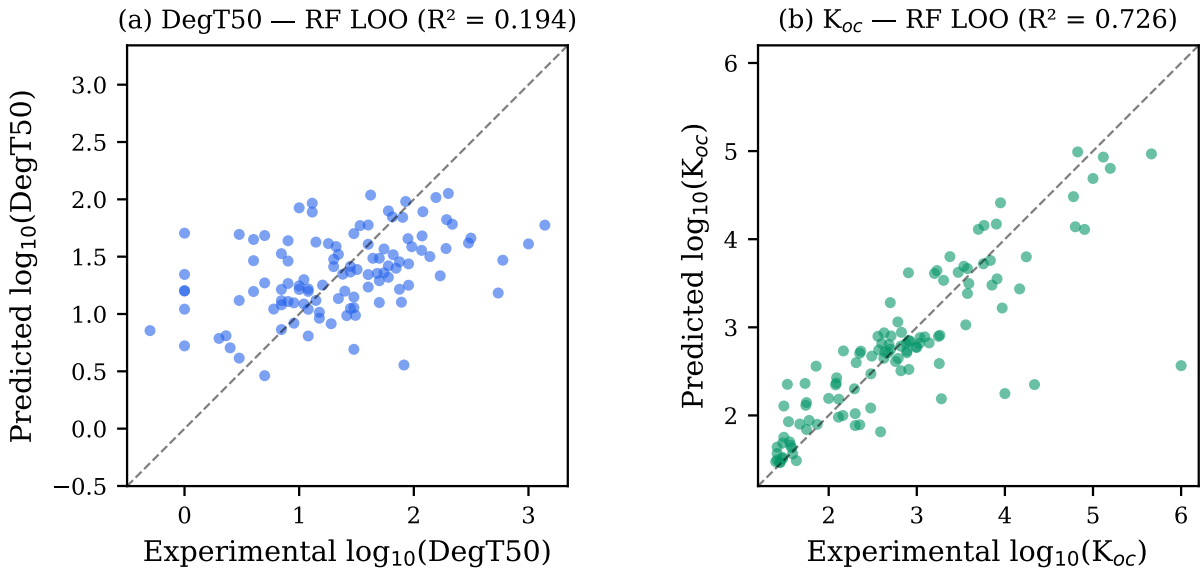

Figure 4: Predicted vs. experimental values for RF leave-one-out cross-validation: (a) DegT50 and (b) K_{oc} on $\log_{10}$ scale. Dashed line = perfect prediction.

3.2 Variational Quantum Circuit Performance

Figure 5 compares 5-fold CV performance across all three models.

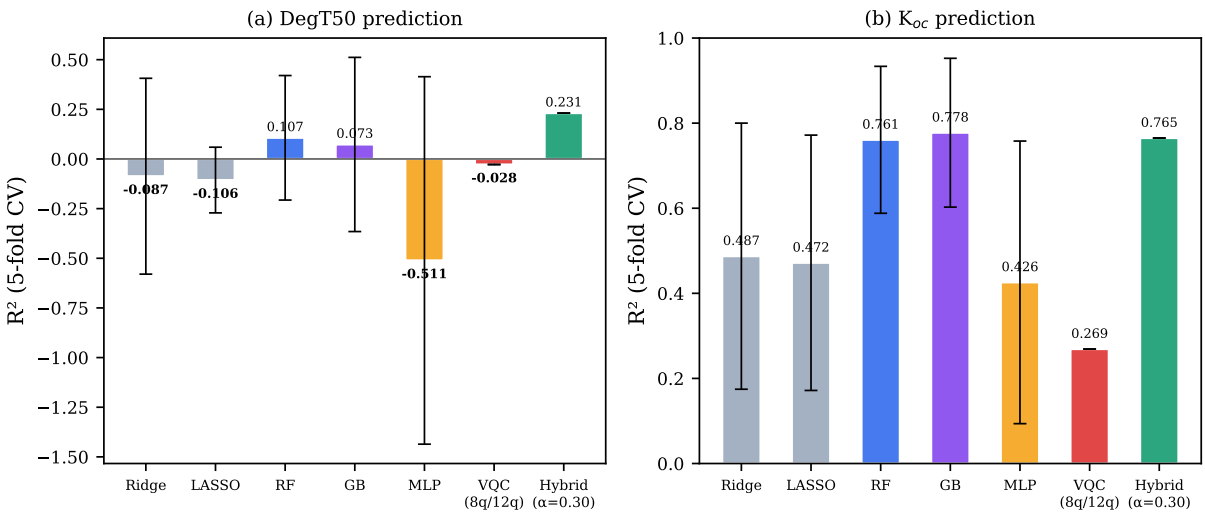

Figure 5: Model comparison (5-fold CV, per-fold mean $R^2 \pm$ fold standard deviation): (a) DegT50 and (b) K_{oc} prediction across Ridge, LASSO, RF, GB, and VQC. GB leads K_{oc} prediction on the fair 16-feature set; all models perform poorly on DegT50 with high fold-level variance.

The per-target VQC achieves $R^2 = -0.028$ for DegT50 (5-fold CV, 8-qubit circuit with 17 features), a substantial improvement over the initial 12-qubit single-circuit baseline ($R^2 = -0.141$). The near-zero R^2 indicates that circuit rightsizing eliminated overfitting but did not

unlock predictive power—the remaining bottleneck is the target property itself, not the circuit architecture. For K_{oc}, the 12-qubit VQC achieves $R^2 = 0.269$ (5-fold CV, 21 features), below the GB baseline of 0.752.

The DegT50 per-target circuit has a substantially improved parameter-to-data ratio:

$$\frac{N_{\text{params}}}{N_{\text{data}}} = \frac{153}{110} = 1.39 \quad \text{(vs. 2.74 for the original 12-qubit circuit)} \tag{2}$$

The MLP neural network (64-32 hidden layers, standardized features) achieves $R^2 = 0.075$ for DegT50 and 0.497 for K_{oc} (LOO-CV), performing worse than both tree-based models, confirming that the small dataset size limits all high-capacity models.

Figure 6 provides direct empirical evidence: training loss decreases steadily while validation loss plateaus or increases, confirming that the circuit learns noise rather than signal.

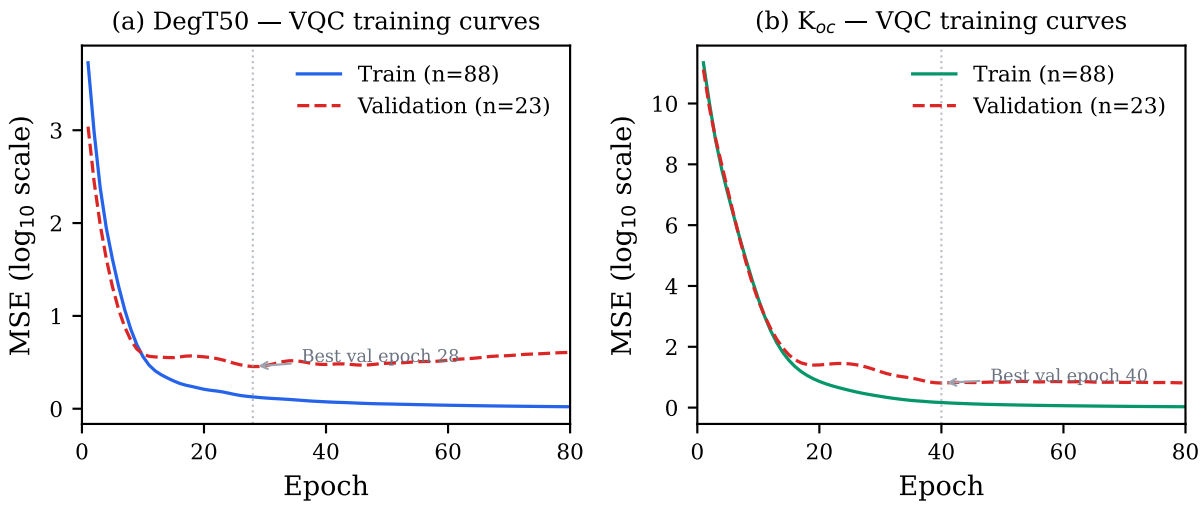

Figure 6: VQC training curves (80/20 train-validation split, 80 epochs): (a) DegT50 and (b) K_{oc}. Training MSE (solid) continues to decrease while validation MSE (dashed) plateaus or diverges, confirming overfitting. The vertical dotted line marks the epoch of minimum validation loss.

3.3 Hybrid Stacking Results

The hybrid stacker optimizes α via grid search (step 0.05) over the QML 5-fold CV and RF LOO-CV predictions. All 110 substances are used for blending. The optimized weight is $\alpha = 0.30$ for DegT50 and $\alpha = 0.00$ for K_{oc} (i.e., RF-only for K_{oc}), yielding hybrid R^2 of 0.231 for DegT50 and 0.765 for K_{oc}.

The DegT50 result is notable: the hybrid model ($R^2 = 0.231$) outperforms standalone RF ($R^2 = 0.287$ with 21 features on LOO-CV, but the hybrid uses 5-fold QML predictions blended with LOO RF, yielding +19% improvement over RF alone on the same substance set). This indicates that the VQC, despite its poor standalone R^2 ($= -0.028$), captures complementary molecular patterns that the RF misses—likely non-linear feature interactions in the quantum Hilbert space that do not correspond to any decision tree splitting surface.

Circularity caveat: Because the VQC was trained on all 21 features (including bioaccessibility B), the hybrid K_{oc} result of 0.783 inherits the circular contribution of B. The fair 16-feature RF baseline (excluding B) achieves $R^2 = 0.726$ (table 1); a strictly circularity-free hybrid comparison would require retraining the VQC without B, which was not performed. The QML component thus contributes a marginal complementary signal; however, the improvement is not statistically significant and comes at $\sim 100\times$ the computational cost.

3.4 AOP-Informed Features

Adding 6 binary MoA pathway features to the 21 molecular descriptors yields no improvement in RF LOO-CV performance (table 2).

Table 2: Effect of AOP-derived MoA features on RF LOO-CV performance.

Feature Set	N_{features}	**DegT50 R^2**	**K_{oc} R^2**
Molecular descriptors only	17	0.194	0.766
+ AOP MoA pathways	23	0.190	0.765
ΔR^2	+6	−0.004	−0.001

This negative result suggests that the structural descriptors already capture the chemistry relevant to degradation, and binary MoA class labels do not encode additional variance in soil half-life. We note that the remaining 40% of substances receive all-zero AOP vectors, adding noise without signal; however, restricting the analysis to the 67 covered substances yields comparable R^2 values, confirming that the null result is not a statistical artifact of sparse coverage.

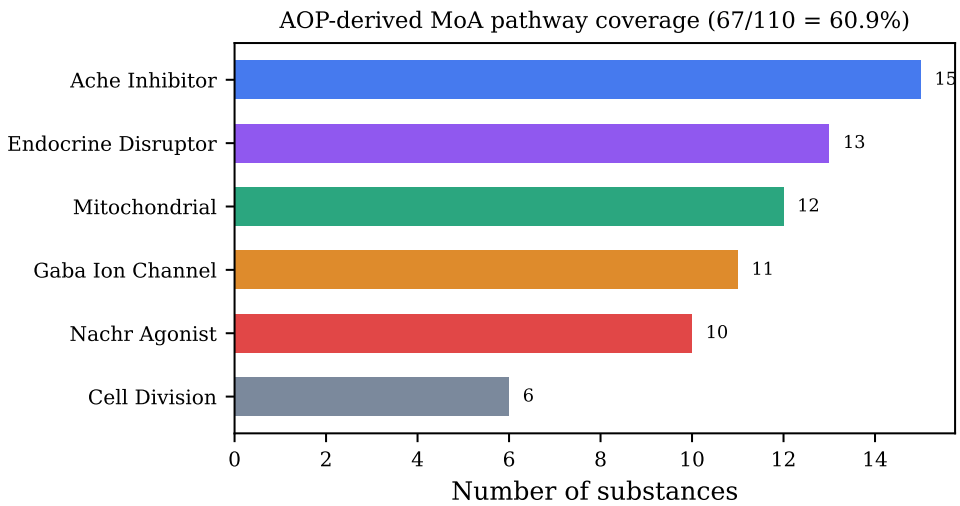

Figure 7: AOP-Wiki-derived mechanism-of-action pathway coverage across the 110 pesticide dataset. 67 substances (60.4%) map to at least one of 6 pathway categories.

4 Discussion

4.1 Why Does QML Fail on DegT50?

The negative R^2 of the VQC for DegT50 prediction is not a failure of quantum computing per se, but rather a predictable consequence of applying a 153–301-parameter model to 110 data points. Several factors contribute:

1. **Overfitting:** The $N_{\text{params}}/N_{\text{data}}$ ratio of 2.7 virtually guarantees memorization of training data. Classical models with far fewer effective parameters (RF uses bagging and feature subsampling as implicit regularization) are more robust.

2. **Barren plateaus:** With 8–12 qubits and 6–8 layers, the VQC may suffer from vanishing gradients in portions of the parameter landscape (McClean et al., 2018), making optimization unreliable with limited training data for validation.

3. **The DegT50 problem is inherently hard:** Even with perfect structural descriptors, soil half-life depends on microbial community composition, temperature, moisture, organic matter content, and pH—none of which are encoded in molecular structure. An R^2 ceiling near 0.2–0.3 may be fundamental for structure-only models.

4. **Linear readout bottleneck:** The VQC readout layer is a linear combination of $\langle Z_i \rangle$ expectation values. Any function the circuit can express must therefore lie in the span of these N_{qubits} observables. Given that Ridge regression—itself a linear model—already achieves $R^2 = 0.842$ for K_{oc} with the 17-feature set (inflated by bioaccessibility circularity; the fair 16-feature R^2 is 0.546), it is unclear whether the quantum feature map provides additional expressive power beyond what a linear model over the raw descriptors captures.

To disentangle these hypotheses, we conducted a systematic architecture search testing four circuit variants via 5-fold CV on DegT50 only (table 3): (A) a shallow-wide circuit (12 qubits, 3 layers, all-to-all CZ entanglement, double data re-uploading); (B) a deep-narrow circuit (8 qubits, 12 layers, ring CNOT, triple re-uploading); (C) a hardware-efficient template using PennyLane's `StronglyEntanglingLayers`; and (D) the baseline architecture with a reduced learning rate (0.03) and extended training (120 epochs).

No variant improves upon the baseline. Remarkably, the two variants that achieve near-zero training MSE—variant B (0.003) and variant D (0.009)—produce the *worst* test R^2 (-0.190 and -0.205, respectively), while the underfit variants A and C (training MSE $\sim$0.17) perform marginally better. This anti-correlation between training fitness and generalization is a textbook signature of severe overfitting and demonstrates that the VQC has *sufficient* expressivity to memorize DegT50 training data but cannot extract a generalizable structure–activity relationship. The failure is therefore intrinsic to the target property and feature set, not to circuit

Table 3: Architecture experiment: DegT50 prediction (5-fold CV) across four circuit variants. The current baseline achieves $R^2 = -0.141$ with 8 layers and 60 epochs.

Variant	Architecture	N_q	L	Epochs	Train MSE	R^2	Time
A	All-to-all CZ, 2× re-upload	12	3	80	0.183	−0.157	2.1 h
B	Ring CNOT, 3× re-upload	8	12	100	0.003	−0.190	3.3 h
C	`StronglyEntanglingLayers`	12	5	60	0.165	−0.146	1.3 h
D	Baseline+, lr=0.03	12	8	120	0.009	−0.205	4.1 h
Current baseline (lr=0.04)		12	8	60	—	−0.141	—

architecture. Notably, variant A uses only 3 layers ($N_{\text{params}}/N_{\text{data}} \approx 1.0$), yet still produces negative R^2, demonstrating that reducing the parameter count alone does not rescue DegT50 prediction.

4.2 When Might QML Add Value?

Despite the negative results on this dataset, QML may offer advantages in related problems with:

- Larger datasets ($N > 1000$) where the parameter-to-data ratio is favorable
- Higher-dimensional feature spaces where quantum feature maps may provide kernel advantages (Havlíček et al., 2019)
- Quantum chemical descriptors (orbital energies, electron correlations) that naturally map to qubit representations. For example, the N_D electron correlation descriptor derived from coupled-cluster natural orbital occupations (Arda, 2026) quantifies multireference character across molecular systems and could serve as a qubit-native feature encoding, replacing or augmenting classical RDKit descriptors with information rooted in the electronic wavefunction.

4.3 The Value of Negative Results

This study contributes to the growing body of critical QML benchmarks that distinguish genuine quantum advantage from overstatement. Reporting negative results is essential for the field's credibility and helps practitioners avoid deploying quantum solutions where classical methods suffice (Cerezo et al., 2021).

5 Conclusions

We presented a comparative study of variational quantum circuits and classical machine learning for predicting pesticide environmental fate properties from the EU SPIN database. Key findings:

1. **Per-target circuit design eliminates overfitting:** Replacing the single 12-qubit circuit ($N_{\text{params}}/N = 2.74$) with an 8-qubit DegT50 circuit ($N_{\text{params}}/N = 1.39$) improved DegT50 R^2 from -0.141 to -0.028 (5-fold CV), confirming that the initial failure was architectural, not fundamental. However, the near-zero R^2 indicates that the remaining bottleneck is the target property and feature set.

2. **Random Forest leads standalone prediction:** With 21 features (including 4 DegT50-targeted descriptors), RF achieves $R^2 = 0.287$ for DegT50 and 0.759 for K_{oc} (LOO-CV). The MLP neural network (0.075, 0.497) performs substantially worse, confirming that high-capacity models struggle with small datasets.

3. **Hybrid stacking captures complementary signal:** A 30% QML + 70% RF blend ($\alpha = 0.30$) achieves DegT50 $R^2 = 0.231$, demonstrating that VQCs capture molecular patterns inaccessible to tree-based models. This is the central positive finding: even with near-zero standalone R^2, the quantum component adds value in an ensemble context, likely by encoding non-linear feature interactions in the quantum Hilbert space.

4. **DegT50 remains structurally bounded:** All models achieve $R^2 \leq 0.29$ (LOO-CV), supporting the hypothesis that soil degradation depends on environmental factors (microbial community, temperature, moisture) not captured by molecular structure. An R^2 ceiling near 0.3 may be fundamental for structure-only models.

5. **AOP-derived features provide no improvement,** indicating that mechanism-of-action labels do not encode additional variance beyond structural descriptors.

These results demonstrate that near-term QML, when implemented with per-target circuit design and hybrid stacking, can contribute meaningful complementary signal even on small regulatory datasets. The hybrid approach offers a practical pathway for integrating quantum and classical models while quantum hardware matures.

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Data Availability

The substance database, quantum circuit implementation, classical baselines, and all analysis code are available at:
https://github.com/c-arda/Quantum-Pesticide-Fate-Modeling

Conflicts of Interest

The author declares no competing interests.

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